Amira Mouamine

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Education

Laval University — Quebec, QC
Bachelor of Engineering in Computer Engineering — *Aug. 2021 – May 2026*
- Relevant Coursework: Game Development Programming, Machine Learning, Real-Time Embedded Systems, Computer Graphics, Database Models and Languages

Technical Skills

Programming Languages: C/C++, C#, Java, Python, JavaScript/TypeScript, Rust, SQL, VHDL, Matlab
Frameworks & Libraries: PyTorch, Tensorflow, Node.js, React.js, Docker, OpenCV, OpenGL, Swing, MySQL
Tools: Git, Azure DevOps, Azure Blob Storage, Blender, Unity, Unreal Engine, Xilinx

Experience

Bentley Systems — May 2025 – Aug. 2025
Full Stack Developer Intern
- Delivered a **Manual Validation feature** unlocking **long-requested flexibility** for civil engineers by enabling persistent measurements previously impossible to save.
- Owned and implemented the **end-to-end measurement system** (create, auto-save, rename, duplicate, delete, recover) across multiple views, significantly **reducing data loss and rework**.
- Integrated secure cloud storage using **Azure Blob Storage (SAS)**, extending existing data persistence mechanisms to manual measurements.

Projects

Graphics Editor | *C++, GLSL, OpenGL, openFrameworks*
- Built support for interactive **3D** scenes (camera controls, lighting, materials, shaders) alongside **2D** drawing tools.
- Enabled image and **3D model import**, **real-time rendering** and instant camera render export for **interactive visualization**.

Sun-Rey Game Jam Project | *C#, Unity, Aseprite*
- Solo-developed a **2D pixel-art game**, handling player movement, collisions, state management and input systems.
- Created and animated original assets (8-direction character animation, tile-based levels) within a solarpunk theme.

Floaty-Frog Game | *Unreal Engine*
- Led a **hands-on tutoring workshop** introducing Unreal Engine through a Flappy Bird-inspired game, guiding participants in implementing core **gameplay mechanics** (movement, collisions, scoring).
- Designed a clean and reusable project architecture using **Blueprints** and **UMG**, tailored for teaching.

Movie Recommendation System | *Python, scikit-learn*
- Built a hybrid recommendation system combining clustering, semantic similarity and supervised quality prediction.
- Processed movie descriptions and genres to recommend content according to user preferences.

Motion Logger | *Rust, OpenCV, Warp*
- Implemented **real-time motion detection** using grayscale frame differencing with **OpenCV**.
- Built a modular system (camera, server, logger) exposing motion data via a **Warp HTTP server**.

Social Engagements

Artificial Intelligence Club of Laval University — Sept. 2025 – May 2026
Member
- Worked on an EMG signal processing and ML-based **gesture recognition** to enable **real-time hand-controlled driving**, progressing from simulation testing to **F1TENTH** deployment with **VAUL**.

Game Development Club of Laval University — Sept. 2024 – May 2026
Co-Leader
- Organized and led the club's activities, including **tutoring workshops**, **industry talks** and **game jams**.
- Represented the Faculty of Science & Engineering at **Eurêka! Festival** in Montreal and the club at the **Girls & Science** event.